HUMSAFAR

A Bi-Directional Remote Intervention and Care Coordination System for Vulnerable Individuals

Project Overview & Reference Guide

Consolidated from SRS v1.0 and SDS v1.0 (Final Locked Baselines)

This document is a quick-reference summary only. The SRS and SDS remain the authoritative sources.

Document Status: REFERENCE — Not a controlling document

Source Documents: Humsafar SRS v1.0; Humsafar SDS v1.0

Department: Computer Science (BSSE)

Team: Muhammad Abdullah (Team Lead), Muhammad Talha Khan, Syed Sarim Jawad Zaidi

Submitting to: S.M. Khalid Jamal

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1. Project Title

Humsafar — A Bi-Directional Remote Intervention and Care Coordination System for Vulnerable Individuals.

2. Executive Summary

Humsafar is a mobile-first care-coordination platform connecting Guardians (Primary and Secondary) with Dependents — vulnerable individuals requiring monitoring and periodic intervention — through layered, largely automated crisis detection, real-time communication, medical record management, and access to external healthcare and home-care services. The system is built as a layered client-server architecture: two native Android applications (Guardian and Dependent) communicate with a backend server over a RESTful API, organized into eight domain-aligned service modules. *(Source: SRS 1.4, 2.1; SDS 5.)*

3. Problem Statement

(Synthesized from the SRS's stated purpose and scope — the SRS does not carry a standalone "Problem Statement" section; this reflects the need the system's requirements were written to address.)

Vulnerable individuals — the elderly, chronically ill, physically disabled, and dementia patients — require continuous care and monitoring that a Guardian cannot always provide in person. The system's requirements reflect a need for layered, largely automated detection of a Dependent's distress — through missed-medication escalation, scheduled Wellness Checks, and Dependent-initiated Panic Mode — rather than relying on any single mechanism, alongside centralized health information and coordinated access to external care services. *(Source: SRS 1.1, 1.4, 2.1.)*

4. Project Objectives

(Derived from the SRS's Purpose, Scope, and Product Functions — not a separately titled SRS section.)

- Provide secure, verified, role-based Guardian-Dependent care coordination.
- Deliver layered, largely automated emergency detection and multi-tier escalation.
- Support real-time text, voice, and video communication between Guardians and Dependents.
- Centralize medical records, health metrics, and passive wearable-based health data.
- Extend access to external healthcare organizations and home-care professionals.
- Maintain accessibility for the target Dependent population through a simplified interaction mode.

5. Scope

In scope *(SRS 1.4):* identity and permission management; Guardian-Dependent relationship management; location tracking and geofencing; medication management; layered emergency detection and escalation; text/voice/video communication; medical records management; wearable health-band integration; a healthcare service directory; a home-care provider directory with request-to-book; on-demand and event-triggered camera viewing; device health monitoring; prioritized notifications; security, encryption, and data-retention controls; accessibility features; a caregiver analytics dashboard.

Out of scope *(SRS 1.4):* auto-answer emergency audio link; continuous 24/7 CCTV recording; in-app payments and formal provider vetting; voice-guided interaction; custom wearable hardware; FHIR/HL7 interoperability; formal HIPAA/GDPR certification; real-time dispatch-API integration; iOS platform support.

6. Target Users and Stakeholders

User Class	Description
Primary Guardian	Registered, verified adult; creates and links Dependent accounts; full permission authority; exclusive access to Remote Alarm Override.
Secondary Guardian	Registered, verified adult invited by a Primary Guardian; access scoped by dashboard permissions.
Dependent	Elderly, chronically ill, physically disabled, or dementia patients requiring continuous care; interacts primarily through Dependent Mode.
Home-Care Professional	Directory-listed nurse/physiotherapist; no platform account in this release.

Project stakeholders: the development team (three BSSE students) and the academic supervisor, S.M. Khalid Jamal. *(Source: SRS 2.3; project cover metadata.)*

7. System Overview

Humsafar follows a layered, modular-monolith client-server architecture: a presentation layer (Guardian and Dependent Android applications), an interface layer (a REST API gateway with JWT authentication), an application layer (eight service modules), and a data layer (PostgreSQL, Redis, and object storage). The system integrates with five categories of external service: a BLE wearable health band, a KYC/liveness identity provider, a push notification service, a real-time communication SDK, and an SMS/voice gateway. *(Source: SDS 5, 6.1.)*

8. Final Feature List (Organized by Module)

Part A — Identity, Access & Guardian-Dependent Relationship

- Role-Based Registration & Guardian Hierarchy (FR-A1)
- Dependent KYC (FR-A2)
- Guardian Identity Verification (FR-A3)
- Guardian Permission Dashboard (FR-A4)
- Guardian-Dependent Linkage (FR-B1)
- Guardian Cap Enforcement (FR-B2)
- Multi-Dependent Support per Guardian (FR-B4)

Part B — Location, Emergency Detection & Escalation, Device Health

- Continuous Adaptive Location Tracking (FR-C1)
- Geofence Definition & Breach Alerts (FR-C2)
- Multi-Zone Geofencing (FR-C3)
- Location History (FR-C4)
- Medication Missed-Dose Escalation (FR-D1)
- Panic Mode (FR-D2)
- Remote Alarm Override (FR-D3)
- Guardian Wellness Check (FR-D4)
- Multi-Level Escalation Chain (FR-D5)
- Fall Detection (FR-D6)
- External Emergency Contact Fallback (FR-R1)
- Battery/Connectivity Monitoring (FR-L1, L2)

Part C — Communication & Notifications

- Text Messaging (FR-E1)
- Voice & Video Calling (FR-E2)
- Priority Voice Messages (FR-E4)
- Multi-Guardian Bridge Call (FR-E5)
- Priority-Tiered Notifications (FR-M1)
- Notification Acknowledgment Tracking (FR-M2)

Part D — Medication, Medical Records & Health Metrics

- Medication Scheduling & Dual Confirmation (FR-F1)
- Medical Vault & PDF Export (FR-G1)
- Field-Level Access Control (FR-G2)
- Doctor-Shareable Export Link (FR-G3)
- Appointment & Event Tracker (FR-G4)
- Manual Daily Vitals Entry (FR-H1)
- Wearable Health Band Integration (FR-H2)

Part E — Healthcare Directory & Home-Care Marketplace

- Healthcare Service Directory (FR-I1)
- Home-Care Provider Directory & Request-to-Book (FR-J1)
- Shared Activity Log (FR-I2/J2)

Part F — Surveillance, Security & Data Management

- On-Demand Live Camera View (FR-K1)
- Event-Triggered Clip Capture (FR-K2)
- End-to-End Encryption (FR-N1)
- Encryption at Rest (FR-N2)
- Sensitive-Action Audit Logging (FR-N3)
- Data Retention & Deletion Policy (FR-O1)

Part G — Accessibility & Analytics

- Dependent Mode / Simplified UI (FR-P1)
- Large Text & High-Contrast Mode (FR-P2)
- Multilingual (Urdu) Support (FR-P4)
- Adherence & Vitals Analytics Dashboard (FR-Q1)
- Automated Summary Reports (FR-Q2)

Note: FR-P3 (Voice-Guided Interaction), FR-E3 (Auto-Answer Emergency Audio), FR-J3 (Payments/Vetting), and FR-K3 (Continuous CCTV) are Future Scope or Removed — see Section 22.

9. Functional Module Breakdown

Module	Responsibility
Identity & Access Module	Registration, KYC/liveness verification, permission dashboard, Guardian-Dependent linkage
Location & Geofence Module	Continuous tracking, geofencing, location history, device battery/connectivity monitoring
Emergency & Escalation Module	Panic Mode, Remote Alarm, Wellness Check, multi-level escalation, external contact, fall detection
Communication Module	Messaging, voice/video calling, priority voice messages, multi-guardian bridge calls
Medication & Records Module	Medication scheduling, medical vault, appointments, vitals, wearable data ingestion
Directory & Marketplace Module	Healthcare directory, home-care provider directory and booking, shared activity log
Security & Surveillance Module	Camera viewing, clip capture, encryption enforcement, audit logging, data retention
Notification Module	Priority-tiered delivery and acknowledgment tracking, used across all modules

(Source: SDS 7.1.)

10. User Roles and Permissions

- **Primary Guardian:** full permission authority over a linked Dependent; exclusive access to Remote Alarm Override; can grant/revoke Secondary Guardian access per module.
- **Secondary Guardian:** access limited to modules explicitly granted — Live Location, Health Records, Communication, Emergency Alert Visibility, Camera Viewing, Medication Management.
- **Dependent:** interacts primarily through Dependent Mode for Panic Mode, Wellness Check response, and Medication Confirmation.
- A Guardian's Primary/Secondary role is fixed at the account level and applies uniformly across every linked Dependent.

(Source: SRS 4.1, FR-A1, FR-A4.)

11. Final List of Actors

Actor	Description
Primary Guardian	Full permission authority; exclusive Remote Alarm Override access.
Secondary Guardian	Access scoped by dashboard permissions.
Dependent	Primary interaction via Dependent Mode.
External Emergency Contact	Non-account-holding last-resort SMS/call recipient.
Home-Care Professional	Directory-listed; no platform account.
Backend System	Automated actor for escalation timing, notification delivery, scheduled triggers.

(Source: SRS 6.1.)

12. Final Use Cases (Summary)

ID	Use Case	Related FR(s)
UC-01	Register as Guardian or Dependent	FR-A1–A3
UC-02	Link Dependent to Primary Guardian	FR-B1, B2
UC-03	Invite Secondary Guardian	FR-B1
UC-04	Configure permission dashboard	FR-A4
UC-05	View live location	FR-C1, C2
UC-06	Configure geofence(s)	FR-C2, C3
UC-07	Configure medication schedule	FR-F1
UC-08	Confirm medication taken	FR-F1
UC-09	Configure Wellness Check schedule	FR-D4
UC-10	Respond to Wellness Check	FR-D4
UC-11	Trigger Panic Mode	FR-D2
UC-12	Trigger Remote Alarm Override	FR-D3
UC-13	View live camera feed	FR-K1, K2
UC-14	Communicate (message/call)	FR-E1, E2, E4
UC-15	Manage medical records	FR-G1–G4
UC-16	Record daily vitals	FR-H1
UC-17	Sync wearable data	FR-H2
UC-18	Browse healthcare directory	FR-I1
UC-19	Request home-care service	FR-J1
UC-20	View analytics dashboard	FR-Q1, Q2
UC-21	Receive emergency escalation	FR-D5, R1
UC-22	View audit log	FR-N3

(Source: SRS 6.2.)

13. Functional Requirements (Summary)

47 requirement groups across 7 parts. Full "shall" statements are in SRS Section 4; this is an ID-and-name index only.

FR ID	Name	Priority
FR-A1	Role-Based Registration & Guardian Hierarchy	Core
FR-A2	Dependent KYC	Core
FR-A3	Guardian Identity Verification	Core
FR-A4	Guardian Permission Dashboard	Core
FR-B1	Guardian-Dependent Linkage	Core
FR-B2	Guardian Cap Enforcement	Core
FR-B4	Multi-Dependent Support per Guardian	Core
FR-C1	Continuous Adaptive Location Tracking	Core
FR-C2	Geofence Definition & Breach Alerts	Core
FR-C3	Multi-Zone Geofencing	Recommended
FR-C4	Location History / Route Playback	Optional
FR-D1	Medication Missed-Dose Escalation	Core
FR-D2	Panic Mode	Core
FR-D3	Remote Alarm Override	Core
FR-D4	Guardian Wellness Check	Core
FR-D5	Multi-Level Escalation Chain	Recommended
FR-D6	Fall Detection	Optional
FR-R1	External Emergency Contact Fallback	Recommended
FR-L1	Battery Monitoring & Low-Battery Alert	Core
FR-L2	Connectivity-Loss Alert	Recommended
FR-E1	Text Messaging	Core
FR-E2	Voice & Video Calling	Core
FR-E4	Priority Voice Messages	Core
FR-E5	Multi-Guardian Bridge Call	Optional
FR-M1	Priority-Tiered Push Notifications	Core
FR-M2	Notification Acknowledgment Tracking	Recommended
FR-F1	Medication Scheduling & Dual Confirmation	Core
FR-G1	Medical Vault & PDF Export	Core

FR-G2	Field-Level Access Control on Records	Recommended
FR-G3	Doctor-Shareable Export Link	Optional
FR-G4	Appointment & Event Tracker	Core
FR-H1	Manual Daily Vitals Entry	Core
FR-H2	Wearable Health Band Integration	Core (Mandated)
FR-I1	Healthcare Service Directory	Core
FR-J1	Home-Care Provider Directory & Request-to-Book	Core
FR-I2/J2	Shared Activity Log	Recommended
FR-K1	On-Demand Live Camera View	Core
FR-K2	Event-Triggered Clip Capture	Recommended
FR-N1	End-to-End Encryption	Core
FR-N2	Encryption at Rest	Core
FR-N3	Sensitive-Action Audit Logging	Recommended
FR-O1	Data Retention & Deletion Policy	Recommended
FR-P1	Dependent Mode (Simplified UI)	Core
FR-P2	Large Text & High-Contrast Mode	Core
FR-P4	Multilingual (Urdu) Support	Core
FR-Q1	Adherence & Vitals Analytics Dashboard	Recommended
FR-Q2	Automated Summary Reports	Optional

14. Non-Functional Requirements (Summary)

NFR ID	Name	Summary
NFR-01	Command Latency	Remote commands initiate within 2s over 4G
NFR-02	Battery Efficiency	Background telemetry ≤5%/hour; excludes active sessions
NFR-03	Communication Confidentiality	E2E encryption for audio/video
NFR-04	Data-at-Rest Confidentiality	Records, metrics, and location encrypted at rest
NFR-05	System Availability	99.9% uptime; local queuing during outages
NFR-06	Scalability	3 concurrent Guardian connections per Dependent
NFR-07	Usability (Dependent Mode)	Safety-critical actions completable within 2 taps
NFR-08	Accessibility Standard	WCAG 2.1 AA contrast guidance (best-practice benchmark)
NFR-09	Localization Completeness	100% English/Urdu coverage
NFR-10	Data Privacy Principles	General data-protection principles; no formal certification claimed
NFR-11	Maintainability	Independent development/testing per module
NFR-12	Portability	Android API 26 through latest stable; iOS out of scope
NFR-13	Wearable Sync Reliability	Auto-reconnect within 30s of returning to range

(Source: SRS Section 5.)

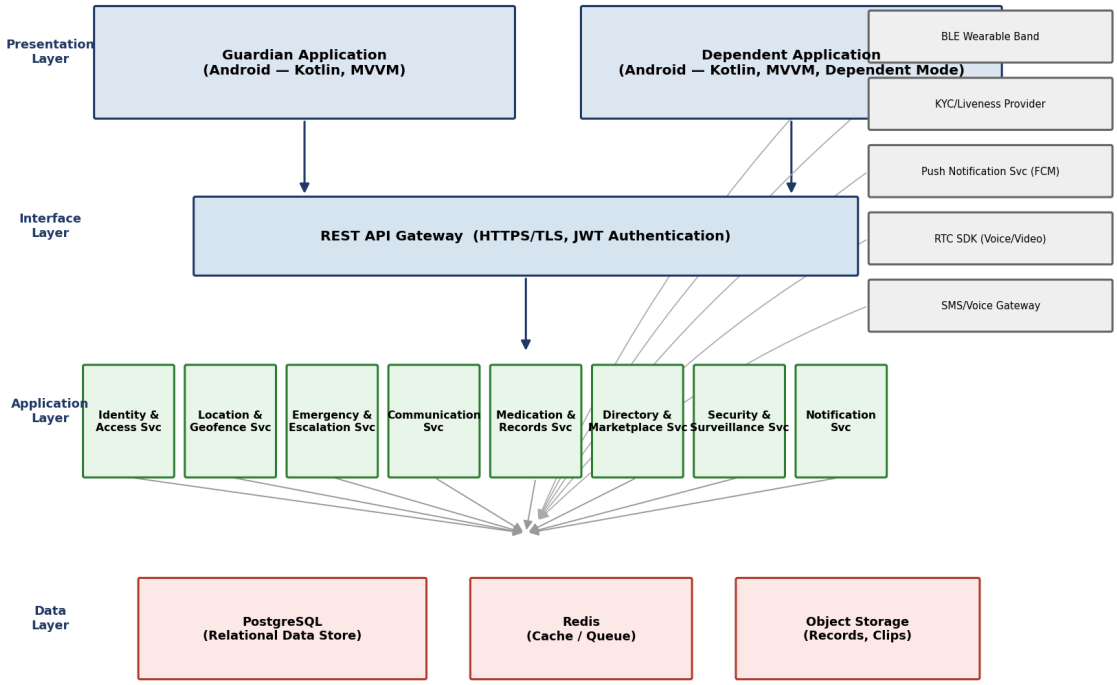
15. Technology Stack (Chosen in the SDS)

Component	Technology
Mobile Client	Kotlin, MVVM architecture, native Android (API 26–latest)
Backend API	Node.js with TypeScript, REST over HTTPS
Primary Database	PostgreSQL (primary + read replica)
Cache / Queue	Redis
Object Storage	Cloud object storage (medical records, camera clips)
Push Notifications	Firebase Cloud Messaging (FCM)
Real-Time Communication	Third-party RTC SDK (voice/video calling)
Identity Verification	Third-party KYC/liveness verification provider
SMS/Voice Gateway	Third-party telephony gateway (External Emergency Contact Fallback)
Wearable Integration	Bluetooth Low Energy, standard GATT profiles
Authentication	JWT (access + refresh tokens)

(Source: SDS 24.1.)

16. High-Level System Architecture Summary

Figure 6.1 — Overall System Architecture (Layered Client-Server)

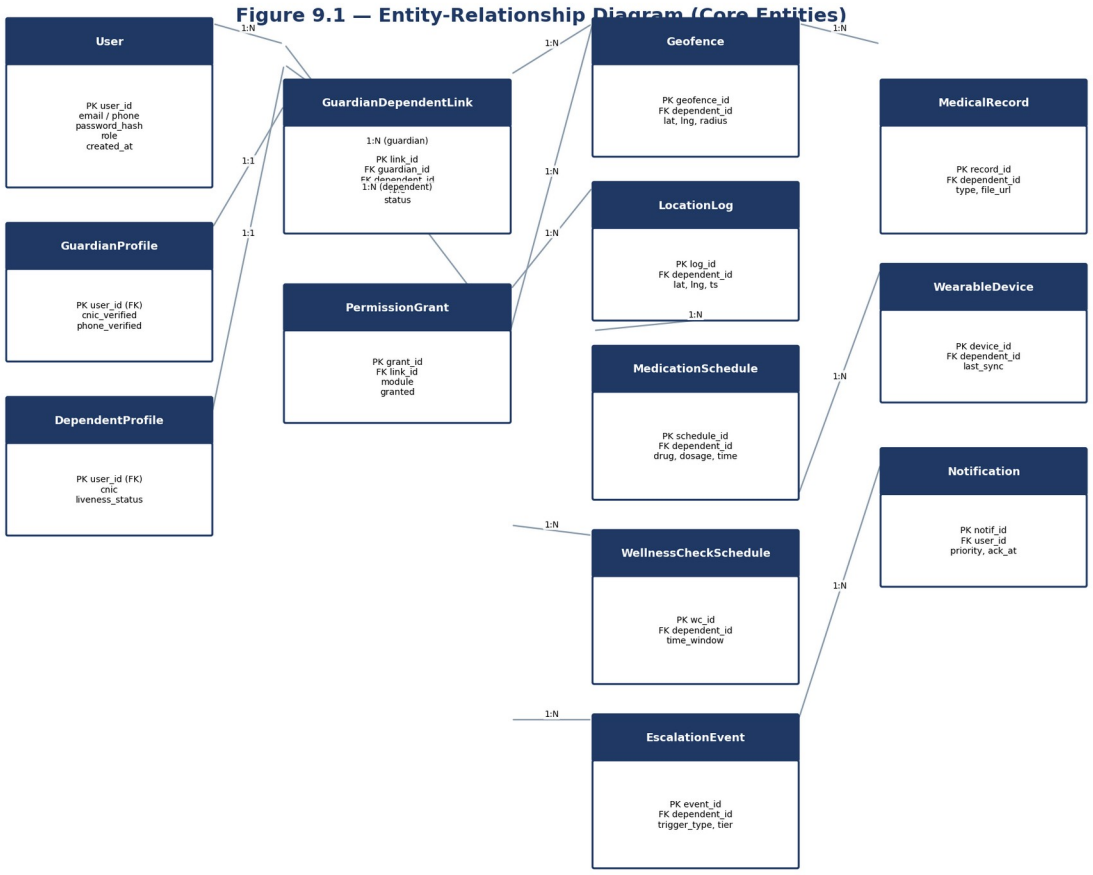


Overall System Architecture (SDS Figure 6.1)

Layered, modular-monolith architecture chosen over microservices, given the three-person team and single-year timeline. *(Source: SDS 6.1, 6.2.)*

17. Database Overview

PostgreSQL, selected for strong ACID guarantees and foreign-key/check-constraint support given the permission- and consent-sensitive nature of the data. **27 tables** in total; twelve core entities shown below. *(Source: SDS 9.1, 9.2.)*



Entity-Relationship Diagram — Core Entities (SDS Figure 9.1)

18. API Overview

RESTful API over HTTPS, secured with stateless JWT authentication (15-minute access tokens with refresh tokens). All permission checks are enforced server-side against the permission-grant table, independent of client UI logic. **26 endpoints** are specified across identity, location, emergency, communication, medication/records, directory, camera, and notification functions. *(Source: SDS 11.)*

Representative endpoints:

- POST /auth/register, POST /auth/login
- POST /dependents/{id}/panic — Panic Mode trigger
- POST /dependents/{id}/alarm — Remote Alarm Override (Primary only)
- POST /dependents/{id}/wellness-checks, POST /wellness-checks/{id}/respond
- GET /dependents/{id}/camera/stream
- POST /wearables/sync

19. Security Features

Authentication & Authorization: Stateless JWT; server-side permission enforcement on every request.

Identity Verification: CNIC + liveness for Dependents; CNIC/OTP for Guardians, via a third-party KYC provider.

Encryption in Transit: TLS 1.2+ for all traffic; end-to-end encryption for audio/video.

Encryption at Rest: Medical records, health metrics, and location history encrypted.

Audit Logging: Permission changes, camera views, and record access logged and viewable by the Primary Guardian.

Camera Access Control: Permission-gated; on-screen indicator during active viewing.

BLE Pairing Security: Standard BLE bonding; GATT data accepted only from the paired device address.

Input Validation & API Hardening: Server-side schema validation, parameterized queries, rate limiting on auth/KYC endpoints.

(Source: SDS Section 17.)

20. Key Design Decisions

Decision	Rationale
Modular monolith over microservices	Matches a three-person team and single-year timeline
PostgreSQL over a document/NoSQL store	Highly relational data benefits from ACID + FK constraints
REST over GraphQL	Simpler tooling and team familiarity within the timeline
JWT (stateless) over server-side sessions	Enables horizontal scaling without shared session storage
Kotlin + MVVM over cross-platform frameworks	Android-only scope removes the main cross-platform benefit; full BLE/background-service access
Redis for cache and lightweight queueing	Reduces infrastructure components for a three-person team to operate
On-screen camera-viewing indicator	Addresses a privacy/consent concern at negligible engineering cost

(Source: SDS Section 22.)

21. Project Constraints and Assumptions

Constraints:

- No offline mode; active internet connectivity assumed as normal operating condition.
- Wearable integrated via standard BLE GATT only; no custom hardware.
- Camera functionality assumes an existing, off-the-shelf IP camera.
- No formal HIPAA/GDPR-equivalent certification claimed.
- Team size (three developers) and single-academic-year timeline.
- English and Urdu localization required; Android-only target.

Assumptions:

- The Dependent's device supports Bluetooth Low Energy.
- A compatible wearable band is obtained outside the application.
- External healthcare organization contact information is publicly available.
- Home-Care Professionals are vetted through means external to the application.
- Health record and metric entry may be performed by either the Dependent or the Guardian.

(Source: SRS 2.5, 2.7.)

22. Future Enhancements (Documented Future Scope)

- Auto-answer emergency audio link (removed from MVP; superseded by Priority Voice Messages).
- Continuous 24/7 CCTV recording and cloud storage.
- In-app payments and a formal home-care provider vetting/ratings pipeline.
- Voice-guided interaction for the Dependent.
- Custom wearable hardware or firmware.
- FHIR/HL7 medical-data interoperability.
- Formal HIPAA/GDPR-equivalent compliance certification.
- Real-time dispatch-system API integration with external healthcare/emergency organizations.
- iOS platform support.

(Source: SRS Section 1.4.)

23. Complete Project Glossary

Term	Definition
Guardian	A registered caregiver account; either Primary or Secondary.
Primary Guardian	The Guardian who created a Dependent's account; full permission authority; exclusive Remote Alarm Override access.
Secondary Guardian	A Guardian invited by a Primary Guardian; access scoped per module.
Dependent	A vulnerable individual monitored and supported through the system.
Dependent Mode	The simplified, large-target navigation mode for safety-critical actions.
KYC	Know Your Customer — identity verification via CNIC and, for Dependents, liveness capture.
Guardian Wellness Check	Scheduled, Guardian-configured prompts requiring Dependent confirmation of well-being.
Panic Mode	Dependent-initiated emergency alert; a supplementary, not primary, detection layer.
Escalation Chain	The multi-tier notification sequence triggered by an unacknowledged critical alert.
BLE / GATT	Bluetooth Low Energy / Generic Attribute Profile — wearable band integration protocol.
Geofence	A Guardian-defined geographic boundary triggering an alert on exit.
Activity Log	The shared log for Healthcare Directory contacts and Home-Care Marketplace requests.
MVVM	Model-View-ViewModel — the client presentation-layer pattern.
REST	Representational State Transfer — the backend API architectural style.
JWT	JSON Web Token — the stateless API authentication token format.
RBAC	Role-Based Access Control.
ERD	Entity-Relationship Diagram.
Out of Scope / Future Scope	See Section 22.

(Source: SRS 7.1; SDS 4.3.)

24. Project Summary

Humsafar is fully specified through two frozen baseline documents: the **SRS v1.0** (47 functional requirement groups, 13 non-functional requirements, 22 use cases, 6 actors) and the **SDS v1.0** (complete architectural, database, API, and interface design, with 100% traceability from every design element back to its governing requirement). This reference guide consolidates both into a single navigable summary and introduces no information beyond what those two documents already establish. Any future change to scope, requirements, or design must be made in the SRS or SDS directly; this document should then be regenerated to match, not edited independently.